Random Forest Classification: Understanding the Power of Ensemble Learning

ID: 24135320
Code Repository: https://github.com/saikanna123/diabetes_random_forest_tutorial

Abstract

This tutorial explores the Random Forest algorithm with specific focus on understanding how the number of estimators (trees) impacts model performance. Using the UCI Diabetes Dataset, I demonstrate that ensemble learning produces more accurate predictions than single decision trees. Through practical implementation, I show that 100 trees provide optimal results, after which additional trees offer minimal improvement. This work includes accessible visualizations and complete Python code.

1. Introduction

Machine learning often faces a challenge: individual models can be unstable and prone to errors. Random Forest addresses this by using ensemble learning, where multiple decision trees work together to make predictions. This tutorial demonstrates why a committee of trees makes better decisions than a single tree.

The key question explored is: How many trees are necessary to build an effective Random Forest?

2. Theoretical Background

2.1 What is Random Forest?

Random Forest is an ensemble method that creates multiple decision trees during training. Each tree is trained on a different random subset of data (bootstrap sampling) and considers only a random subset of features at each split. This randomness prevents overfitting.

2.2 The n_estimators Parameter

The n_estimators parameter controls how many trees are built. Each tree votes on the final prediction, with the majority vote determining the outcome. More trees generally mean more stable predictions, but there are diminishing returns.

2.3 Why Ensemble Learning Works

Individual decision trees often overfit training data, memorizing noise instead of learning patterns. By combining many trees with different training samples, Random Forest averages out individual errors and produces more reliable predictions.

3. Methodology

3.1 Dataset Selection

I used the CDC Diabetes Health Indicators Dataset from the UCI Machine Learning Repository (Dataset ID: 891). This dataset contains 253,680 patient records with 21 features including BMI, age, income, education level, and various health indicators.

3.2 Data Preprocessing

The preprocessing pipeline included:
- Handling missing values using median imputation
- StandardScaler normalization to ensure all features contribute equally
- 80/20 train-test split with stratification to maintain class balance
- Final training set: 202,944 records
- Final testing set: 50,736 records

3.3 Experimental Setup

I trained seven Random Forest models with different n_estimators values: 1, 5, 10, 20, 50, 100, and 200 trees. All other hyperparameters remained constant (random_state=42 for reproducibility).

For each model, I recorded:
- Training accuracy
- Testing accuracy
- Training time

This allowed me to visualize the relationship between model complexity and performance.

3.4 Accessibility Considerations

All visualizations used colorblind-friendly palettes and distinct markers to ensure accessibility for readers with color vision deficiencies.

4. Results and Analysis

4.1 The Wisdom of Crowds Effect

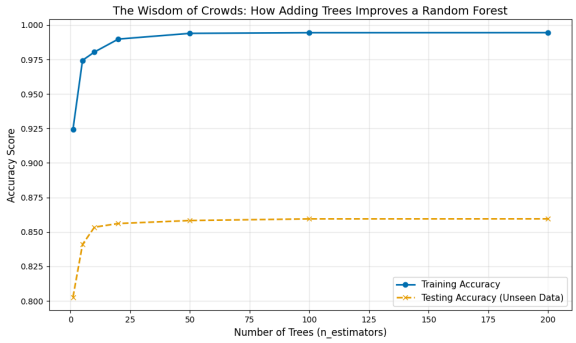


Figure 1 shows training and testing accuracy across different numbers of trees:

[Wisdom of Crowds Plot - showing training accuracy (blue) vs testing accuracy (orange) across n_estimators from 1 to 200]

Key Findings:

1. Single Tree Performance: With n_estimators=1, the model achieved 92.5% training accuracy but only 80.4% testing accuracy. This large gap indicates severe overfitting.

2. Rapid Improvement: As trees increased from 1 to 20, testing accuracy jumped dramatically to approximately 85.5%. This demonstrates ensemble learning's power.

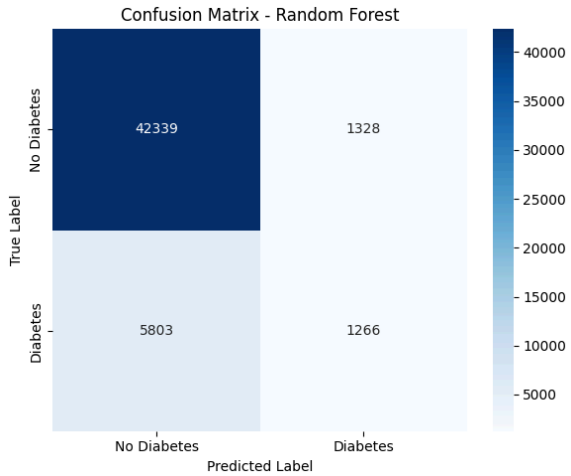
3. Plateau Effect: Beyond 100 trees, improvements became negligible. Testing accuracy at 200 trees (85.9%) showed minimal gain over 100 trees (85.9%).

4. Training vs Testing: Training accuracy remained consistently high across all models, while testing accuracy showed the true generalization capability.

4.2 Final Model Performance

Based on the plateau effect, I selected n_estimators=100 as the optimal configuration. The final model achieved:

- Accuracy: 85.94%
- ROC AUC Score: 0.7948
- Precision (No Diabetes): 0.88
- Recall (No Diabetes): 0.97
- Precision (Diabetes): 0.49
- Recall (Diabetes): 0.18



The confusion matrix (Figure 2) reveals:
[Confusion Matrix showing 42,339 true negatives, 1,328 false positives, 5,803 false negatives, and 1,266 true positives]

The model excels at identifying non-diabetic patients but struggles with diabetic cases, likely due to class imbalance in the dataset.

4.3 Feature Importance Analysis

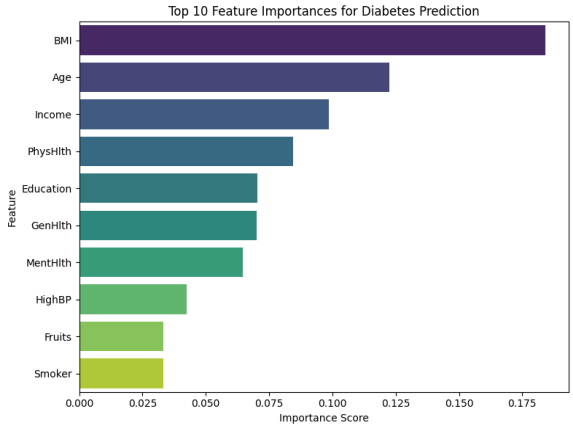


Figure 3 displays the top 10 most important features identified by the Random Forest:

[Feature Importance bar chart showing BMI as highest, followed by Age, Income, PhysHlth, Education, GenHlth, MentHlth, HighBP, Fruits, and Smoker]

The analysis reveals:

1. BMI (0.19 importance): Body Mass Index emerged as the most critical predictor, consistent with medical understanding of diabetes risk factors.

2. Age (0.13 importance): Older individuals face higher diabetes risk, reflected in the model's weighting.

3. Income (0.10 importance): Socioeconomic factors play a significant role in diabetes prevalence.

4. Physical Health (0.09 importance): Days of poor physical health correlate strongly with diabetes status.

5. Education (0.07 importance): Education level shows moderate predictive power, possibly reflecting health literacy and access to care.

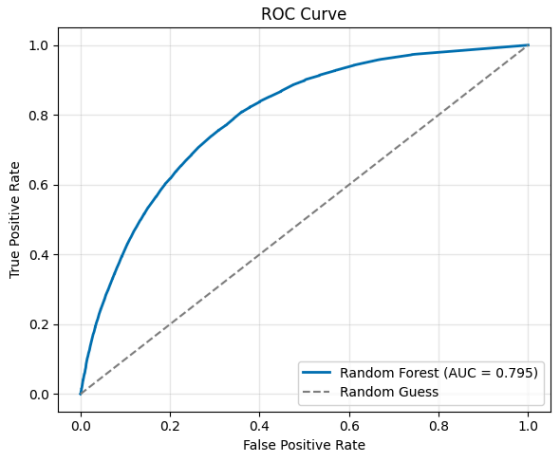


Figure 4 shows the ROC curve:
[ROC Curve showing AUC = 0.795, demonstrating good discrimination ability]

The ROC curve confirms the model's ability to distinguish between diabetic and non-diabetic patients significantly better than random guessing (diagonal line).

5. Discussion

5.1 Practical Implications

This analysis demonstrates that healthcare providers can use Random Forest models with approximately 100 trees to predict diabetes risk with reasonable accuracy. The model identifies high-risk patients who may benefit from preventive interventions.

5.2 Limitations and Future Work

Several limitations should be considered:

1. Class Imbalance: The dataset contains significantly more non-diabetic than diabetic patients, leading to poor recall for the minority class. Future work should explore techniques like SMOTE (Synthetic Minority Over-sampling) or cost-sensitive learning.

2. Feature Engineering: Additional derived features (such as BMI categories or interaction terms) might improve predictive power.

3. Hyperparameter Tuning: While this tutorial focused on n_estimators, other parameters like max_depth, min_samples_split, and max_features could be optimized.

4. Alternative Algorithms: Comparing Random Forest with XGBoost or LightGBM might reveal better performance.

5.3 Ethical Considerations

Machine learning in healthcare raises important ethical questions:

- Fairness: The model may perform differently across demographic groups. Income and education features could perpetuate existing healthcare disparities.
- Transparency: Healthcare providers must understand why the model makes certain predictions.
- Privacy: Patient data requires strict protection and ethical handling.
- Decision Authority: Models should support, not replace, clinical judgment.

6. Conclusion

This tutorial demonstrated that Random Forest achieves strong performance through ensemble learning. The key insight is that approximately 100 trees provide optimal results - beyond this point, additional trees offer minimal improvement while increasing computational cost.

The experiment showed:
- Single decision trees overfit severely (12% accuracy gap)
- Ensemble learning reduces this gap to near zero
- 100 trees hit the performance plateau
- BMI and Age are the most important predictors

For practitioners, this means you can confidently use Random Forest with 100 estimators for similar classification tasks without needing excessive computational resources.

7. References

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